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|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Pepsin</b> | Gastric Chief Cells  | Proteolytic (protein-digesting) enzyme that cleaves long protein chains into smaller peptides and peptones.                    |
| <b>Mucus</b>  | Gastric Goblet Cells | Forms a protective gel layer over the inner stomach epithelium, preventing self-digestion and corrosion from concentrated HCl. |

*Chyme & Pyloric Sphincter:* Partially digested acidic semi-solid food in the stomach is called **chyme**. Small quantities of chyme are released periodically into the small intestine through the regulated muscle ring called the **pyloric sphincter**.

## 4. Small Intestine: Complete Digestion & Absorption

The **small intestine** is the longest part of the alimentary canal and the site of complete digestion of carbohydrates, proteins, and fats. Herbivores have longer small intestines to allow cellulose digestion, whereas carnivores have shorter ones.

### A. Liver & Gallbladder Contributions (Bile)

- **Bile Secretion:** The liver secretes green-yellow **bile**, which is stored temporarily in the **gallbladder**.
- **Alkaline Conversion:** Acidic chyme arriving from the stomach is neutralized and made alkaline by bile, creating the ideal pH for pancreatic enzymes.
- **Emulsification of Fats:**

## 5. Large Intestine, Reabsorption & Egestion

Unabsorbed and undigested material moves into the **large intestine** (colon). The walls of the colon absorb remaining water and essential salts from the waste material, transforming it into solid waste (feces). Feces are stored temporarily in the **rectum** and ultimately eliminated from the body through the **anus** under the regulation of the *anal sphincter*.

## 6. Key Reference Matrix: Digestive Enzymes & End Products

Organ / Gland

